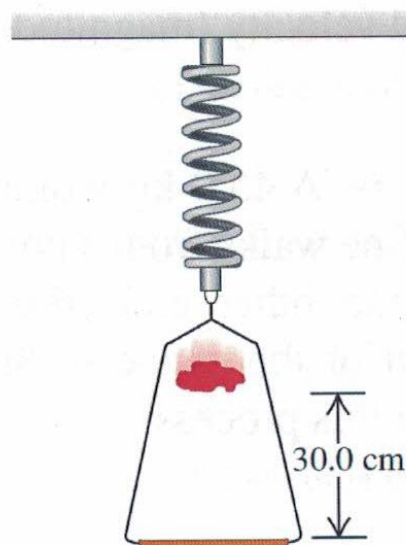


Tutorial 4
Momentum, Impulse & Collision

Q8.9 In a completely inelastic collision between two objects, where the objects stick together after the collision, is it possible for the final kinetic energy of the system to be zero? If so, give an example in which this would occur. If the final kinetic energy is zero, what must the initial momentum of the system be? Is the initial kinetic energy of the system zero? Explain.

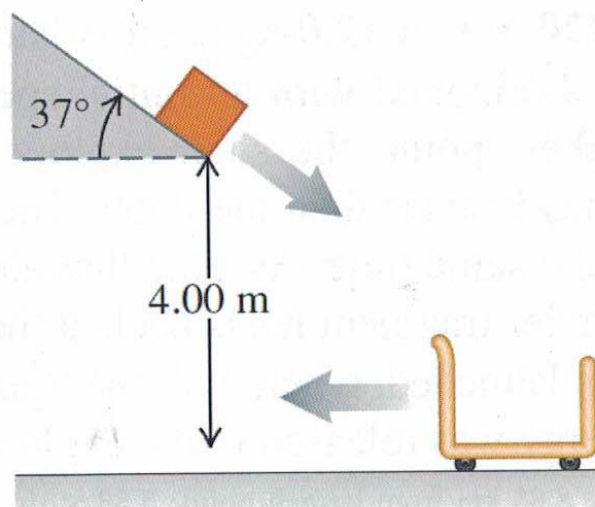
8.82 ... CP A 0.150-kg frame, when suspended from a coil spring, stretches the spring 0.070 m. A 0.200-kg lump of putty is dropped from rest onto the frame from a height of 30.0 cm (Fig. P8.82). Find the maximum distance the frame moves downward from its initial position.

Ans: 29.5 cm



8.95 .. CP In a shipping company distribution center, an open cart of mass 50.0 kg is rolling to the left at a speed of 5.00 m/s (Fig. P8.95). You can ignore friction between the cart and the floor. A 15.0-kg package slides down a chute that is inclined at 37° from the horizontal and leaves the end of the chute with a speed of 3.00 m/s. The package lands in the cart and they roll off together. If the lower end of the chute is a vertical distance of 4.00 m above the bottom of of

Figure **P8.95**



the cart, what are (a) the speed of the package just before it lands in the cart and (b) the final speed of the cart?

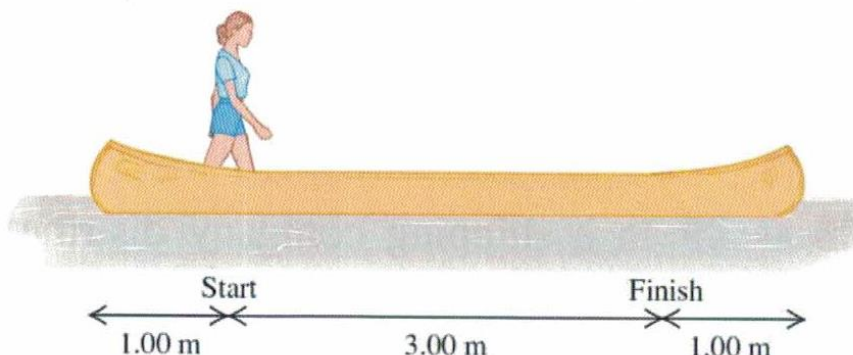
Ans: (a) 9.35 m/s (b) 3.29 m/s to the left

8.104 •• Jonathan and Jane are sitting in a sleigh that is at rest on frictionless ice. Jonathan's weight is 800 N, Jane's weight is 600 N, and that of the sleigh is 1000 N. They see a poisonous spider on the floor of the sleigh and immediately jump off. Jonathan jumps to the left with a velocity of 5.00 m/s at 30.0° above the horizontal (relative to the ice), and Jane jumps to the right at 7.00 m/s at 36.9° above the horizontal (relative to the ice). Calculate the sleigh's horizontal velocity (magnitude and direction) after they jump out.

Ans: 0.105 m/s to the right

8.106 •• A 45.0-kg woman stands up in a 60.0-kg canoe 5.00 m long. She walks from a point 1.00 m from one end to a point 1.00 m from the other end (Fig. P8.106). If you ignore resistance to motion of the canoe in the water, how far does the canoe move during this process?

Ans: 1.29 m to the left

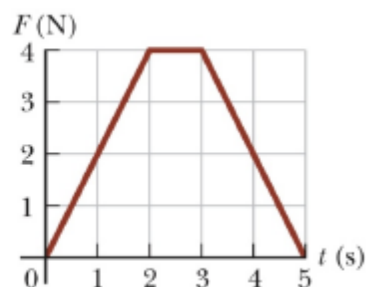


8.110 ••• A 12.0-kg shell is launched at an angle of 55.0° above the horizontal with an initial speed of 150 m/s. When it is at its highest point, the shell explodes into two fragments, one three times heavier than the other. The two fragments reach the ground at the same time. Assume that air resistance can be ignored. If the heavier fragment lands back at the same point from which the shell was launched, where will the lighter fragment land, and how much energy was released in the explosion?

Ans: 8.63 km from launch point; 5.33×10^5 J

7

[S9.15] The magnitude of the net force exerted in the x direction on a 2.50-kg particle varies in time as shown in the figure below. Find (a) the impulse of the force over the 5.00-s time interval, (b) the final velocity the particle attains if it is originally at rest, (c) its final velocity if its original velocity is $-2.00 \hat{i}$ m/s, and (d) the average force exerted on the particle for the time interval between 0 and 5.00 s.



Answer: (a) $\vec{I} = 12.0 \text{ N s}$ (b) $\vec{v}_f = 4.80 \text{ m/s } \hat{i}$; (c) $\vec{v}_f = 2.80 \text{ m/s } \hat{i}$; $\vec{F}_{avg} = 2.40 \text{ N } \hat{i}$

8

[G9.32] (a) Calculate the impulse experienced when a 65-kg person lands on firm ground after jumping from a height of 3.0 m. (b) Estimate the average force exerted on the person's feet by the ground if the landing is stiff-legged, and again (c) with bent legs. With stiff legs, assume the body moves 1.0 cm during impact, and when the legs are bent, about 50 cm. [Hint: The average net force on her which is related to impulse, is the vector sum of gravity and the force exerted by the ground.]

Answer (a) 499 Ns (b) 301 mg (c) 7 mg

9

A neutron collides elastically with a helium nucleus (at rest initially) whose mass is four times that of the neutron. The helium nucleus is observed to move off at an angle $\theta'_{He} = 40^\circ$. Determine the angle of the neutron, θ'_n and the speeds of the two particles, v'_n and v'_{He} , after the collision. The neutron's initial speed is $6.2 \times 10^5 \text{ m/s}$.

Answer $v'_{He} = 1.899 \times 10^5 \text{ m/s}$; $v'_n = 4.899 \times 10^5 \text{ m/s}$; $\theta'_n = 85.6^\circ$

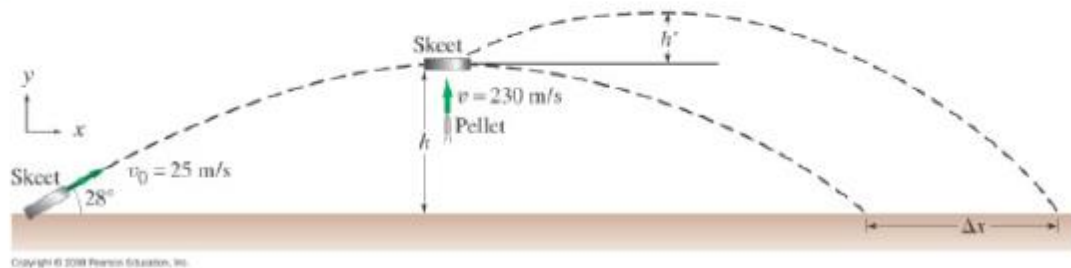
10

[G9.59] A neon atom ($m=20.0 \text{ u}$) makes a perfectly elastic collision with another atom at rest. After the impact, the neon atom travels away at a 55.6° angle from its original direction and the unknown atom travels away at a -50.0° angle. What is the mass (in u) of the unknown atom? [Hint: You could use the law of sines.]

Answer 39.9 u

11

[G9.103] A 0.25-kg skeet (clay target) is fired at an angle of 28° to the ground with a speed of 25 m/s. When it reaches the maximum height, h , it is hit from below by a 15-g pellet travelling vertically upwards at a speed of 230 m/s. The pellet is embedded in the skeet. (a) How much higher h' does the skeet go up? (b) How much extra distance Δx does the skeet travel because of the collision?



Answer (a) 8.6 m (b) 38.4 m

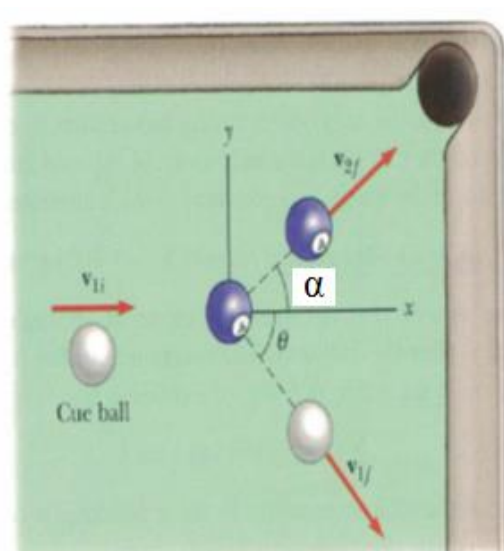
12 Some interesting results about elastic collision of the same mass.

(i) One dimensional collision

A ball travelling with speed u_1 collides elastically head on with another ball of the same mass travelling in the same direction with speed u_2 with ($u_2 < u_1$). Determine the speed of the two balls v_1 and v_2 after the collision.

(ii) Two dimensional collision

In a game of billiards, a player wishes to sink a target ball in the corner pocket as show in the figure below. If the angle to the corner pocket is α , at what angle θ is the cue ball deflected? Assume that friction and rotational motion are unimportant and that the collision is elastic.



13 Oblique Collision

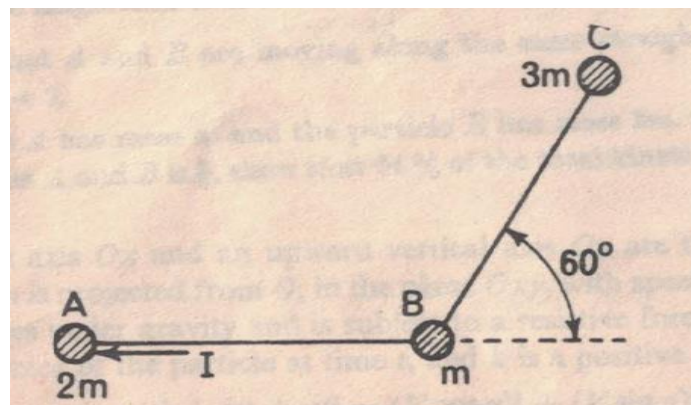
Two smooth spheres A and B, of equal radius but of masses m and M , are free to move on a horizontal table. A is projected with speed u towards B which is at rest. On impact, the line joining their centres is inclined at an angle θ to the velocity of A before impact. If e is the coefficient of restitution between the spheres, find the speed with which B begins to move. If A's path after impact is perpendicular to its path before impact, show that

$$\tan^2 \theta = \frac{eM - m}{M + m}$$

Answer $\frac{(1+e)mu \cos \theta}{M+m}$

14 Three connected particles

The diagram shows three particles A, B and C of mass $2m$, m and $3m$ respectively, initially at rest, joined by two light inelastic strings AB and BC. The angle ABC is 120° . An external impulse I is applied to the mass A in the direction of BA. Determine the velocities with which the three masses begin to move.



Answer

A begins to move with $\frac{13I}{42m}$ along AB.

B begins to move with $\frac{I}{3m}$ at $\tan^{-1}\left(\frac{3\sqrt{3}}{13}\right)$ with BA.

C begins to move with $\frac{I}{21m}$ along CB.